1 PSEUDOCODE

Algorithm 1 trains the embeddings of one domain, Algorithm 2 runs the full experiment (split, embeddings, 12 mappings, evaluation), Algorithm 3 trains one mapping, and Algorithm 4 recommends books to a cold-start user. Notation follows Section 2.

Algorithm 1 TrainEmbeddings

Input: rating triples $D = \{(u, i, r)\}$, dimension k, λ, learning rate α, batch size s, epochs E, seed
Output: user matrix P, item matrix Q

set random seed; $P[u], Q[i] \sim \mathcal{N}(0, 0.01)^k$ for all users u, items i
for $e = 1$ **to** E **do**
  shuffle D
  for each mini-batch $S \subseteq D$ of size s **do**
    $\Delta P, \Delta Q \leftarrow 0$
    for $(u, i, r) \in S$ **do**
      $err \leftarrow r - P[u] \cdot Q[i]$
      $\Delta P[u] \mathrel{+}= err\, Q[i] - \lambda P[u]$
      $\Delta Q[i] \mathrel{+}= err\, P[u] - \lambda Q[i]$
    end for
    $P \leftarrow P + \alpha\, \Delta P / |S|$; $Q \leftarrow Q + \alpha\, \Delta Q / |S|$
  end for
end for
return P, Q

Algorithm 2 RunExperiment

Input: M', B', T=1, fractions $\{0.1, 0.2, 0.5, 1\}$, K, seed
Output: NDCG@n of each strategy at each fraction

$\mathcal{O} \leftarrow \{u : H_b(u) \neq \emptyset \wedge H_m(u) \neq \emptyset\}$
$\mathcal{O}_{test} \leftarrow$ random 20% of $\mathcal{O}$; $\mathcal{O}_{train} \leftarrow \mathcal{O} \setminus \mathcal{O}_{test}$
$truth[u] \leftarrow$ book ratings of u in B' for $u \in \mathcal{O}_{test}$
remove all ratings of $\mathcal{O}_{test}$ from B' ▷ before any training
$(U', V') \leftarrow$ TrainEmbeddings$(M', k, \dots)$
$(U'', V'') \leftarrow$ TrainEmbeddings$(B', k, \dots)$
for each fraction ρ **do**
  $\mathcal{O}_\rho \leftarrow$ sample $\lfloor \rho |\mathcal{O}_{train}| \rfloor$ users from $\mathcal{O}_{train}$ without replacement
  $X \leftarrow \{U'[u]\}_{u \in \mathcal{O}_\rho}$; $Y \leftarrow \{U''[u]\}_{u \in \mathcal{O}_\rho}$
  $g_{shared} \leftarrow$ TrainMapping(g_θ, X, Y)
  $c_{1..K}, C_{1..K} \leftarrow$ KMeans(X, K)
  $g_j \leftarrow$ TrainMapping$(g_{\theta_j}, X_{C_j}, Y_{C_j})$ for $j = 1..K$
  $m_\phi \leftarrow$ TrainMapping$(m_\phi(x)\, x,\ X, Y)$
  for each strategy $g \in \{g_{shared}, g_{cluster}, g_{specific}\}$ **do**
    $score_g(\rho) \leftarrow$ mean over $u \in \mathcal{O}_{test}$ of NDCG@n of Recommend(u, g) against $truth[u]$
  end for
end for
return all $score_g(\rho)$

Algorithm 3 TrainMapping

Input: model g_θ (an MLP, or m_ϕ with $g(x) = m_\phi(x)x$), inputs X, targets Y, α, batch size s, epochs E
Output: trained parameters θ

initialize θ randomly (fixed seed)
for $e = 1$ **to** E **do**
  for each mini-batch S of pairs (x, y) from (X, Y) **do**
    $L \leftarrow \frac{1}{|S|} \sum_{(x,y) \in S} \| g_\theta(x) - y \|^2$
    $\theta \leftarrow \theta - \alpha \nabla_\theta L$ ▷ backpropagation
  end for
end for
return θ

Algorithm 4 Recommend

Input: user u, strategy g, threshold T, list length N
Output: ranked list of N books

if $|H_b(u)| < T$ **then** ▷ cold start
  $x \leftarrow U'[u]$
  if $g = g_{shared}$ **then**
    $e \leftarrow g_{shared}(x)$
  else if $g = g_{cluster}$ **then**
    $j^* \leftarrow \arg\min_j \| x - c_j \|$; $e \leftarrow g_{j^*}(x)$
  else
    $e \leftarrow m_\phi(x)\, x$
  end if
else
  $e \leftarrow U''[u]$
end if
$score[b] \leftarrow e \cdot V''[b]$ for every candidate book $b \notin H_b(u)$
return the N books with the highest $score[b]$